

Toward an Improved Understanding of Agile Project Governance: A Systematic Literature Review

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Abstract

The purpose of this study is to provide understanding of project governance practices in agile projects. A systematic review of the previous agile literature is conducted to identify and categorize agile project governance practices, which are synthesized in a six-dimensional framework. The study contributes to the literature by contrasting traditional and agile project governance practices and by identifying new practices emerging in agile projects, while also noting remaining gaps in knowledge. The implications of this study enable managers to identify and apply feasible governance practices that can support better agile project performance.

Keywords

agile project governance, agile approaches to project management, project governance, practices, systematic literature review

Introduction

Project governance has recently become the topic of intense research efforts; a growing consensus agrees that proper governance is crucial to ensuring the effectiveness of projects and project-based organizations (Ahola, Ruuska, Artto, & Kujala, 2014; Müller, 2009; Müller & Lecoivre, 2014). In the project management literature, project governance is usually viewed as the organizational governance of a project, the system through which an organization controls, directs, and steers the project toward its objectives (McGrath & Whitty, 2015). Up to now, project governance research has mostly been conducted in the context of traditional projects developed in a linear fashion, and in project-based industries that rely on formal planning and control, such as infrastructure and construction projects (Reve & Levitt, 1984; Sanderson, 2012; Turner, 2006). Prior research on project governance has also focused on fixed, higher-level organizational arrangements and procedures, including compliance assurance or business case formation: elements of particular importance during the project front-end phase (Ahola et al., 2014; Müller, Pemsel, & Shao, 2015; Samset & Volden, 2016). However, few studies have investigated actual governance practices at the level of single projects during their entire life cycle. More important, the implications of the flexible project management that is highlighted in contemporary business operations—such as agile projects in the software development industry—for

actual project governance practices have not been widely discussed.

Although agile practices have been developed and studied primarily in the field of software engineering, more recently several other industries and communities have shown increased interest in agile development (Conforto, Rebentisch, & Amaral, 2014a). A global survey by Conforto et al. (2014a) illustrates that although software industry practitioners (37% of the sample of 856) most frequently reported the use of agile practices (88%), use was also reported in financial services (15% of the sample; 36% agile use), consulting (10%; 24%), and even the generally considered traditional metal manufacturing (3%; 8%) and defense industries (2%; 7%). The prevalence of agile practices in different industrial and organizational settings emphasizes the importance of understanding agile governance or, more specifically, agile project

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governance. Instead of formalized planning and control approaches, agile approaches aim to embrace change and uncertainty by relying on highly iterative development of a project's goals and content and by emphasizing informal collaboration (Dybå & Dingsøyr, 2008). Previous research on governance in agile projects and organizations has primarily focused on assessing the maturity of agility or agile transitions in an organization (Conboy, 2009; Gandomani & Nafchi, 2015; van Oosterhout, Waarts, & van Hillergersberg, 2006) and has been approached from an organizational or functional governance perspective—for example, with respect to IT governance (Janssen & van der Voort, 2016; Vlietland, Van Solingen, & Van Vliet, 2016), offering limited empirical evidence. However, to date, actual theorizing on how traditional project governance practices operate differently in the agile project context, and what kinds of novel practices agile projects may require, has been highly limited. Understanding how governance techniques are enacted in agile projects holds the promise to make them even more effective.

The purpose of this study is to increase the understanding of how agile projects are governed. We address this aim with the following research questions: What are the project governance practices applied in agile projects, and how are they different from those used in more traditional projects? This study identifies and categorizes the project governance-related practices found in agile projects in relation to the project governance framework proposed by Kujala, Aaltonen, and Gotcheva (2016), a framework derived from work on traditional projects without attention to different project types or contexts. The present study therefore aims to test whether this framework is complete and how it might apply to agile projects. The study also aims to compare agile and more traditional project governance practices to illustrate the similarities and differences between these approaches, highlighting the novel or missing elements. The study is conducted as a systematic literature review of agile project studies published in software engineering, computer science, and project management journals. Through the integration of Kujala et al.'s (2016) framework and evidence from the literature, this study conceptualizes agile project governance by describing and categorizing governance practices applied in agile projects.

The study contributes to the project governance literature by applying the underlying techniques suggested by Kujala et al. (2016) to the literature on agile approaches to project management, describing the traditional governance practices that can be applied to agile projects and identifying emerging practices in agile environments that differ from traditional project governance, while noting remaining gaps in knowledge. Second, the study isolates those elements related to project governance from the literature on agile approaches to project management, making this theme explicit. The study therefore offers insight for both project management and software engineering disciplines while encouraging further research on this topic in related contexts. Managers across industries will also benefit from the insights offered by this study, suggesting how project governance can be adjusted according to agile characteristics

and how this can impact the organization's performance and its transition toward agility.

Theoretical Background

The Concept of Project Governance and Governance Approaches

Project governance is a multifaceted concept whose inherent nature, conceptualizations, theoretical foundations, and key characteristics have been actively debated by project management scholars (Biesenthal & Wilden, 2014; Müller, 2009). Project governance has been defined, for example, as “the alignment of project objectives with the strategy of the larger organization” (Project Management Institute, 2013, p. 553); “the framework within which project decisions are made” (Garland, 2009, p. 10); or as a model that—based on principal-agent theory—illustrates the roles, responsibilities, and interaction hierarchy of the main project network constituents: project funder, steering committee, project owner, project manager, and project team (Zwikael & Smyrk, 2015). In the existing literature, the mechanisms and practices of project governance have been treated through the higher-level concepts of control in agency theory (Eisenhardt, 1989), trust in stewardship theory (Davis, Schoorman, & Donaldson, 1997), and relationships and relational contracting (Müller, Zhai, Wang, & Shao, 2016), which leaves room for more detailed micro-level examinations of actual governance practices enacted in projects' daily lives.

Two distinct approaches to the governance of projects have been described in the literature: an external view, emphasizing strategy and how the project fits into the organization, and subsequently, a more internal view regarding how projects are managed internally.

The external approach to project governance (Ahola et al., 2014) views governance as “the system by which a project is directed and controlled and held to account” (McGrath & Whitty, 2015, p. 781). This type of governance ensures that the outcomes of the project are aligned with the larger strategy of the organization while also ensuring that the standards and rules that apply to projects are followed in practice (McGrath & Whitty, 2015; Müller & Martinsuo, 2015). In this view, project governance is an embedded part of the project's environment, which has a moderating effect on the project's methodology and success (Joslin & Müller, 2016).

More recently, another tendency has emerged, approaching project governance from an internal perspective (Ahola et al., 2014). In this view, the governance of a single project is examined from the perspective of internal coordination, with a focus on ways to safeguard, coordinate, and adapt the interactions among the participants in the temporary project to ensure that they are working toward shared project goals. The distinction between project management and project governance, therefore, becomes critical. While project management focuses on rational planning and technical implementation of a project from the perspective of central actors, project governance can

be considered a subset of project management aiming to obtain information about the project's current status to enable steering and appropriate intervention and to ensure that the central coordination mechanisms and rules aligning the goals of project actors are followed in practice (Kujala et al., 2016).

After conducting a systematic review of project governance in the project management literature, Kujala et al. (2016) suggested a framework for project governance, identifying its key dimensions and mechanisms (Table 1). The studies Kujala et al. (2016) reviewed generally dealt with projects that applied traditional, regulated, planning-oriented project management methodologies, along with stage gate-based models. Consequently, the findings of the review represented the key governance mechanisms used in traditional projects, not in agile contexts.

Kujala et al. (2016) report six dimensions, where each dimension represents an independent aspect of project governance:

1. The goal setting dimension concerns consolidating business or other requirements and technical and organizational capabilities within the operational frame and project objectives.
2. The incentives dimension aligns the project team and stakeholders' reward and risk-sharing schemas with the project targets.
3. The monitoring dimension includes practical tools and mechanisms for measuring the project's performance.
4. The coordination dimension deals with arrangements that steer the project's future actions and communicate decisions among the parties involved.
5. The roles and decision-making dimension are important for assigning the right people to the right roles with the right responsibilities and powers that support the project's performance toward its objectives. And finally,
6. The capability building dimension involves the mechanisms that help identify and enhance the critical competencies required for projects and further develop them based on ongoing and past project experiences.

Kujala et al.'s (2016) study on project governance is predominantly focused on governance mechanisms enacted in traditional project contexts; therefore, there is room for research that examines whether the mechanisms identified also apply in the context of agile projects. Such an approach can also contribute to the elaboration of a contingency perspective to project governance, which is crucial for understanding what combinations of project governance mechanisms are effective in specific contexts.

Agile Approaches and Projects

The cornerstone for understanding agile approaches and agile approaches to project management is the *Agile Manifesto*, which presented four statements and 12 principles to improve the software development process (Beck et al., 2001). The

Agile Manifesto clearly states that most value exists in individuals and in interactions, while acknowledging that some value also inheres in processes and tools in software development. The *Agile Manifesto* states that "working software [is valued] over comprehensive documentation," "customer collaboration [is valued] over contract negotiation," and "responding to change [is valued] over following a plan" (Beck et al., 2001, p. 2). Consequently, project success should be determined primarily by customer satisfaction and the value the project provides to the customer: "Our highest priority is to satisfy the customer through early and continuous delivery of valuable software" (Beck et al., 2001, p. 1).

The 12 agile principles—for example, "Simplicity—the art of maximizing the amount of work not done—is essential" (Beck et al., 2001, p. 3)—do not entirely contradict the project governance literature, but they clearly encourage continuously asking whether existing project governance mechanisms, processes, and tools are sufficient and effective for delivering value to the customer. The main conceptual differences between traditional and agile software development are described in Table 2.

Ideally, empowered and cross-functional agile teams are solely responsible for determining the best ways of working for each project. For example, Schwaber and Sutherland (2016) describe teams that use the Scrum method as follows:

The Scrum Team consists of a Product Owner, the Development Team, and a scrum master. Scrum Teams are self-organizing and cross-functional. Self-organizing teams choose how best to accomplish their work, rather than being directed by others outside the team. Cross-functional teams have all competencies needed to accomplish the work without depending on others not part of the team. The team model in Scrum is designed to optimize flexibility, creativity, and productivity (p. 5).

The Scrum team is also responsible for integrating the flow of deliverables to achieve the project's desired output goal, and can thus be described as "the core project team" in Chiochio's typology (2015) of project teams. However, in a large agile project, several Scrum teams could work simultaneously, in which case they would be "component project teams" according to the above-mentioned typology.

The literature on agile approaches to project management (Highsmith, 2009) has proposed different approaches for software-intensive products and services development. These approaches include agile value incorporation, agile project scaling, and enhancing organizational agility, and may be closely connected to project governance issues, generating implications that have not yet been fully recognized within the literature on project governance.

Research Methodology

The aim of this study is to improve the understanding of how agile projects are governed. We selected a systematic literature review (Tranfield, Denyer, & Smart, 2003) as the optimal

Table 1. Key Dimensions and Mechanisms of Project Governance Based on a Systematic Literature Review Within the Project Management Field (Modified from Kujala, Aaltonen, & Gotcheva, 2016)

Key Dimensions and Mechanisms of Project Network Governance	Description of Governance Mechanism
Goal setting	
Joint performance goals	High reliance on a contract as a legally binding document that ensures contractually determined outcomes or outputs are delivered on schedule, within budget, and as accepted behavior Early relationship-building meetings and workshops with key stakeholders can be used to agree on schedule planning and objectives In addition, long-term issues need to be considered in strategic decision making concerning the project Conflicts over project objectives or lack of commitment are the main problems
Clarity of goals	Inherent goal orientation of actors Challenge of achieving relevance and sustainability if project objectives are unknown or misunderstood
Flexibility of goals	Front-end plan-driven approaches; flexibility for goals low Rendezvous clauses: revisiting parts of a contract by officially deferring decision making on which agreement cannot be reached at the outset Flexibility in bid requirements to allow contractors to bring their skills and propose innovative solutions Sequential tendering of work to allow changes at a later stage
Incentives	
Rewards tied to performance (short-term)	In performance-based contracts, contractually defined incentives and penalties for defaults Bonuses for completing project ahead of schedule or fines for being late Using target cost and pain/gain share contract to drive performance Integrated project teams with financial incentives to stimulate innovation
Risk allocation	Choice of contract forms (e.g., lump-sum, cost-plus) can shift risk exposure between parties Risk/reward regime based on monetized key performance indicators Traditional risk-sharing approaches particularly in turnkey projects that bring up controversies between the project parties
Rewards tied to life cycle performance (long term)	Use of life cycle approach in which project parties are paid by performance and availability of project product Reductions in payment due to performance (e.g., failure to maintain service standards will result in payment deductions or financial damages)
Ownership structure	Ownership structure/share in arrangements Creating a sense of ownership for participating organizations and individuals (e.g., by retaining an ownership stake in the asset) Continuity of personnel
Reputation and future business	Publishing reputation scoring for individual organizations Reputation as an incentive to sacrifice short-term interests in exchange for long-term interests
Monitoring	
Formal control and monitoring	Behavioral and outcome control in terms of budget, time, and scope Contractually specified monitoring and reporting procedures, sequential approach Contracts setting out comprehensive performance measurement system, including key performance indicators Only realistic and monitored project milestones and performance targets considered useful Periodic progress monitoring Systems and mechanisms for monitoring, controlling, and reporting progress of the work Regular client inspections and site visits Steering group to oversee overall progress and to provide guidance
Third-party monitoring and auditing	Use of external parties to monitor performance Auditing mechanisms
Coordination	
Common project management practices	High reliance on contracts: Contractually specified key principles and agreements among parties (e.g., budget, delivery deadline, quality standards, safety requirements) Strategy document outlining the project vision and organizational processes that will enable it to be achieved Formalized relationships such as boards and group structures that allow a degree of control over required outcomes
Shared culture, values, and norms	A shared set of values, objectives, and beliefs about how to coordinate the organization's efforts to reach common objectives A consciously designed project culture across key participating organizations

(continued)

Table 1. (continued)

Key Dimensions and Mechanisms of Project Network Governance	Description of Governance Mechanism
Communication and information sharing	- Relational norms that define expected behavior - Joint organizational development and training strategy to ensure that cultural change issues are planned and delivered effectively - Regular meetings with project participants to confirm target schedules - Meetings to facilitate communication and decision making - Information dissemination and communication systems - Lines of communication - Information transparency
Change management	- Contractual terms associated with specified principles, tactics, organization structures, and processes for resolving unforeseeable events - Communicating risk using a top-down approach or directly among project participants
Conflict resolution	- Flexible partnerships enabling changes to run smoothly - Court injunction to settle disputes legally - Out-of-court negotiations to settle disputes - Personal relationships between actors to resolve conflicts effectively - Formal statement of values as a basis for resolving disputes internally - Collaborative problem resolution methodology: a systematic approach with realistic timescales for resolutions
Roles and decision making	
Role definition	- Contractually defined roles and responsibilities for each party - Organizational structures
Management structure	- Creation of a suitable project management structure: establishing reporting lines to the top-level project board - Roles and responsibilities of boards and management - Clear governance structure, in which the work management team is responsible for daily execution of the project
Authority for decision making	- Equality between parties to create true partnerships - Right decision at the right time: a form of active participation - Competence and risk-carrying capacity as basis for allocating responsibility - Delegation of power to project team
Capability building	
Actor	- Selecting people with experience and high-quality performance record - Identifying necessary skills and expertise early and ensuring adequate attention to resourcing of project teams - Learning lessons and recruiting managers capable of applying experience gained on other projects
Training and continuous learning	- Tendering process to encourage innovation (optimized contractor involvement, “early enough”) - Periodic learning - Providing training to suppliers (e.g., safety culture) - Systematic collaboration and practice development - Exchange of innovation success and learning from failures - Identifying, articulating, and codifying innovative practices for the future primarily at the end of the project

method because increased understanding requires the examination and interpretation of current theories in the field (Gregor, 2006). As Webster and Watson (2002) suggest, a literature review should aim to contribute to theoretical development by synthesizing the literature on a mature topic or elaborating new concepts in an emerging topic. This method fits the intentions of this study because there is considerable discussion of governance in traditional project management literature, but much less in the research on agile approaches in software engineering. Accordingly, we apply existing theoretical concepts regarding project governance to the context of the agile literature, to identify and make explicit the areas of discussion

that apply to governance and thus contribute to theory development.

This research started with a planning phase (Rivard, 2014) in which the authors formulated the research problem, identified the literature review as the appropriate research protocol, and agreed on the scope and limitations of the study. The overall research process drew on Rowe's (2014) model that provided the following steps for such a literature review: (1) selecting structural dimensions and categories driven by theory, (2) collecting material, (3) evaluating the material using systematic coding and performing synthesis, and (4) discussing and developing the theory. The first step suggested

Table 2. Traditional Versus Agile Software Development (Nerur, Mahapatra, & Mangalaraj, 2005)

	Traditional Development	Agile Development
Fundamental assumptions	Systems are fully specifiable, predictable, and can be built through meticulous and extensive planning	High-quality, adaptive software can be developed by small teams using the principles of continuous design improvement and testing based on rapid feedback and change
Control	Process centric	People centric
Management style	Command and control	Leadership and collaboration
Knowledge management	Explicit	Tacit
Communication	Formal	Informal
Customer's role	Important	Critical
Project cycle	Guided by tasks and activities	Guided by product features
Development model	Life cycle model (waterfall, spiral, or some variation)	The evolutionary-delivery model
Desired organizational form/structure	Mechanistic (bureaucratic with high formalization)	Organic (flexible and participative encouraging cooperative social action)
Technology	No restriction	Favors object-oriented technology

choosing or constructing a theory-driven framework to be applied as a lens in the study. The results of this step were presented in the theoretical background section (see Table 1). To provide a systematic and transparent account of the research process (Rowe, 2014), the stages (Tranfield et al., 2003) and steps followed are illustrated in Figure 1 and described in detail below.

Material Collection

Source Journal Selection. Collecting the material started by selecting sources. Because the project is interdisciplinary and aims to synthesize current, relevant, and respected studies, the sources had to represent the top-ranking journals in the project management and software engineering research streams.

The initial pool of journals was retrieved in October 2016 from the Journal Citation Reports (JCR) of Thomson Reuters Web of Science (WoS) using the following criteria: The journal must be ranked within the top 25 in the management (the category applied to project management) or software engineering (the category applied to agile) categories. Its impact factor must be higher than 1.5 (to be considered good quality). Journals with thematic mismatches (specified topics or methods outside the scope of this article), such as *IEEE Communications Surveys and Tutorials* and the *Journal of Supply Chain Management*, were excluded. These criteria resulted in 19 journals from the JCR.

Another selection of journals was acquired through SCImago Journal and country rank (SJR) using the following criteria: The journal must be within the Q1 rank among the business and international management category (for project management, under the management subject area) or the software category (applied to agile approaches, under the computer science subject area). The SJR figure must be higher than 1.5 (to be considered good quality), and the publication type must be "journal." Journals with thematic mismatches, such as the *International Journal of Robotics Research* and the *Journal of*

Statistical Software, were excluded. These criteria resulted in 22 journals from SJR.

Duplicate entries in the databases, such as the *Academy of Management Annals*, were removed. To support the quality criteria of comprehensive sources (Rowe, 2014), the following journals offering rich subject-matter research and appropriate rigor were added by the consensus of the team of authors: *International Journal of Project Management* and *Project Management Journal* (in project management) and *Journal of Systems and Software* and *Science of Computer Programming* (in software engineering).

Finally, to ensure the rigor and quality of the review process, the journals were searched for "Refereed/peer reviewed" status in Ulrichsweb, since this database provides the journal status and review process information. Ulrichsweb also provided the publication status of the journal, which was determined to be "active." Therefore, the total number of journals was 37, with a total of 174,183 citable items available; see Appendix 1.

Paper Selection. The publications in the selected journals were accessible through the WoS and Scopus databases. The former is generally the most comprehensive, while the latter provides access to the most recent publications because of its more rapid update process. As the search was directed at providing information about project governance mechanisms in the context of agile projects, the papers had to provide content specifically regarding agile projects. Accepted paper types included journal articles and conference proceedings; many practitioners in the agile field prefer to publish their latest contributions in conference proceedings.

The first set of articles was acquired from the WoS database using the advanced search function. The search was filtered to apply to the selected journals, document type, and topic. The detailed search strings are presented in Appendix 2. This procedure resulted in 110 hits, which were exported to a shared Mendeley software database. We also tested the results by adding "GOVERN*" in the topic (TS) field, but this narrowed down the results to only four papers, which is an insufficient

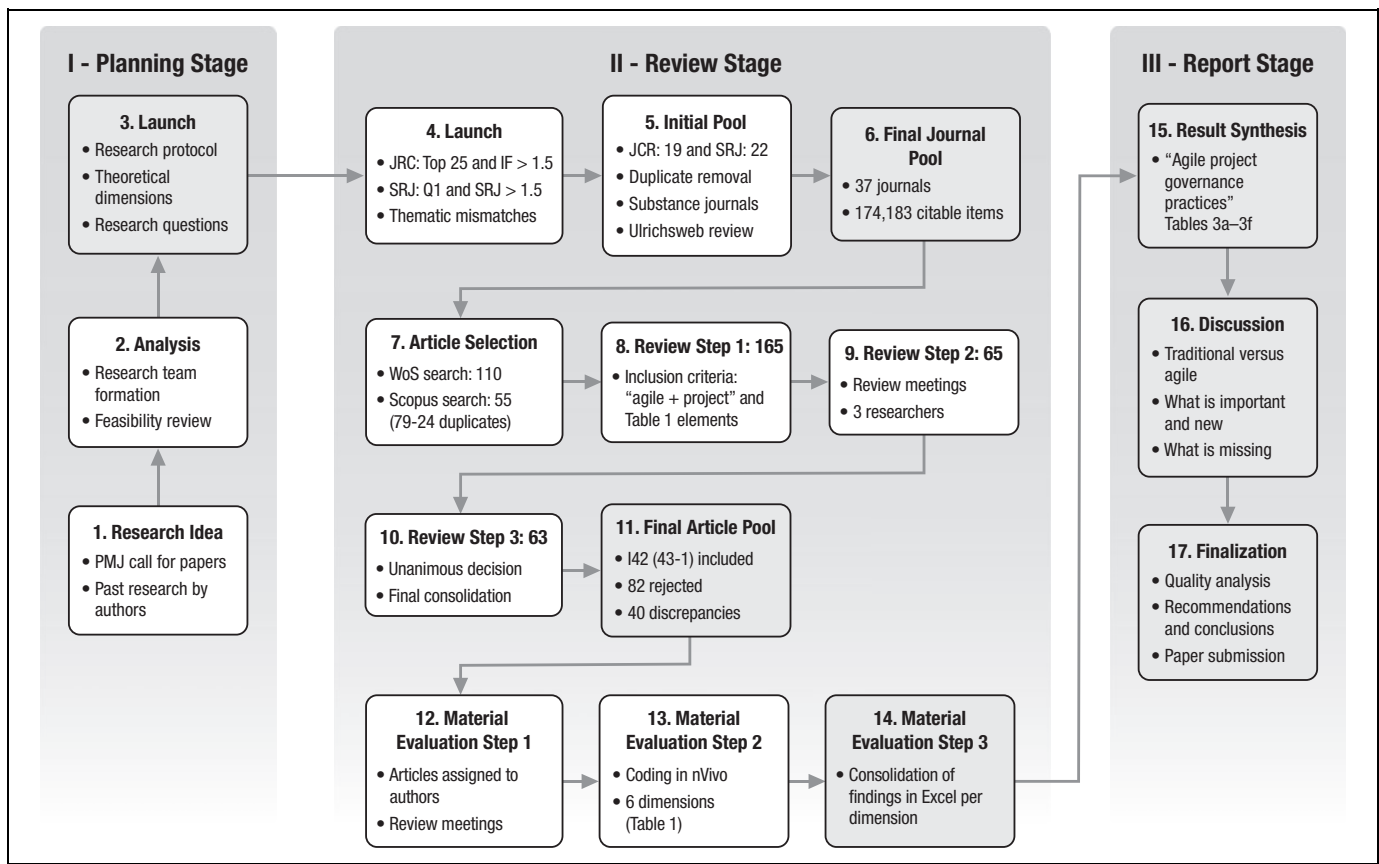


Figure 1. Research process.

pool for a valid literature review. This low number suggested that governance is a novel concept in the agile field, which can be interpreted as support for the rationale of this study.

The second set of articles was acquired from the Scopus database. Because the search tool in Scopus does not provide filtering by publication title, we began by applying only the topic filter “agile project*” to the article title, abstract, and keywords. This yielded 2,374 documents that had to further be filtered by journal title and document type, as illustrated in detail in Appendix 2. The Scopus database also includes publications that are “in press.” Therefore, the document type included the “ip” selection. The result was 79 papers, which were also exported to Mendeley, after which 24 duplicates were removed, leaving a final pool of 165 papers (110 from WoS, 55 from Scopus).

In the next phase, we selected the papers that discussed agile projects in relation to any of the governance framework elements (Tranfield et al., 2003). Although the inclusion criteria for the papers specified projects, many papers discussed agile issues that are outside of this scope, such as agile enterprise transformation. Therefore, the papers were exported from the Mendeley database to Microsoft Excel and were blind reviewed in two steps by three authors (R1–R3) based on the content of their abstracts. The criterion for inclusion was that the abstract must discuss agile projects as well as

several of the governance dimensions and mechanisms presented in Table 1.

The selection was conducted in three review steps, as illustrated in Figure 1. In the first step, a blind review resulted in 65 papers being accepted by at least one author. The results of the first step were reviewed in a joint meeting, to synchronize the inclusion criteria and perspectives among reviewers. This was followed by a second blind review, after which the number of papers accepted by at least one author decreased to 63. In the third and final review step (“10. Review Step 3: 63” in Figure 1), the reviewers agreed that the decision to include a paper in the final pool had to be unanimous. The final pool of articles resulted in 43 papers being unanimously included, 82 unanimously rejected, and 40 papers featuring discrepancies being left out because unanimity was not reached. The details of this inclusion process are presented in Appendix 3. The final pool of articles included one duplicate that had not been identified by Mendeley. This was dropped, leaving a final total of 42 accepted papers. These papers are presented with full references in Appendix 4.

Material Evaluation With Systematic Coding

The material was evaluated qualitatively using nVivo 11 data analysis software. The selected papers were divided among the

authors, who coded the papers individually. The coding unit consisted of text portions in the articles that described governance practices found in Table 1. We looked for these text elements in the original contributions by the source articles' authors, and the elements were coded into the six dimensions of Table 1, or "nodes" as they were presented in nVivo. In several cases, the commonalities, variations, and ambiguity in the textual evidence made it difficult to assign a text element into just one particular governance dimension, which resulted in some being coded to more than one dimension. The coding process itself and the abovementioned ambiguous coding samples were piloted and reviewed in several joint sessions to reach a common understanding of how certain agile practices could be interpreted in relation to the governance dimensions. Finally, the coded elements were synthesized by assigning one or two dimensions to each author, to improve consistency within the dimensions. The results of this analysis and synthesis are presented in the next section.

Agile Project Governance Practices

The results are presented by first synthesizing the key governance practices under each dimension in Table 3. An in-depth analysis of the synthesized findings is provided in the following sections.

Goal Setting

Goal setting includes the project front-end activities that define the project's goal, purpose, strategic alignment, and business case. The main discussion concerning the elements under the goal-setting dimension includes the close cooperation between customers and the empowered project team (Conforto, Salum, Amaral, da Silva, & de Almeida, 2014b), which is critical to reach a shared understanding of the project goals, requirements, and deliverable product vision. Additional actors identified as participating in this project front end, or the explicit "project launch" (Torrecilla-Salinas, Seden, Escalona, & Mejias, 2015), were the stakeholders, technical architects, end users, and usability designers. An agile-specific mechanism mentioned frequently for project goal setting was the use of a product vision, which included the initial, prioritized product features. This vision would be updated as the project progressed, and also influenced the sprint and user story-level goal-setting mechanisms—namely, the minimum viable product (MVP) for "must-have" features and definition of done (DOD) for the acceptance criteria (Augustine, Payne, Sencindiver, & Woodcock, 2005; Daneva et al., 2013; Misra, Kumar, & Kumar, 2009). It was generally accepted in the papers that project objectives could not be explicitly defined up front, but rather evolved during the project life cycle. However, the budgeting process, which is generally an elemental part of goal setting, was mentioned in only four articles and described only as "fluid and short term" in one paper (Conboy & Morgan, 2011, p. 540). Other formal mechanisms associated with the

setting of project goals are contractual arrangements. Contracts were discussed in five articles, with the normative consensus that a flexible, billed-hour type of contract is preferred in agile projects, although customer commitment can be a challenge. Interestingly, only one paper (Stettina & Horz, 2015) discussed how project objectives should align with the company strategy, architecture, and other ongoing project goals.

Incentives

The discussion on the role of incentives was limited in the papers analyzed. Most of the research shares the underlying assumption that the nature and philosophy of agile work itself are the best incentives to get the agile team members committed to project work and to motivate them. For example, Tessem (2014) argues that developers are driven by the possibility of peer recognition, making decisions concerning their tasks, being able to participate in decision making, and having direct access to customers. Consequently, the authors of the studies analyzed have not highlighted the role of monetary or individual incentives but rather the role of recognition in establishing team dedication. Conforto et al. (2014b) also note that organizational arrangements may have significant implications for team members' commitment, suggesting that project-oriented organizational structures support a team's dedication. Although Chow and Cao (2008) proclaim that agile projects can be performed on schedule and under budget even without agile-friendly organizational factors, such as cooperative culture, oral culture, universal acceptance of agile approaches, and an appropriate reward system, Sheffield and Lemetayer (2013) propose that team members' dedication to agile approaches is a key issue and should be nurtured by the parent organization. Evidence concerning risks and particularly the sharing of risks among organizations participating in agile project delivery is also highly limited in the papers selected. None of this research discussed business or commercial models that would support optimizing the overall goals of agile project deliveries in a multi-organizational setup. Interestingly, risk management is discussed primarily from a traditional risk management perspective; that is, relying on planning and control to mitigate risks.

Monitoring

The most common monitoring practices for agile projects, described in the selected papers, were sprint/iteration reviews, in which the main goal was to test and validate deliverables against user story specifications, using automation when possible (Bjarnason, Unterkalmsteiner, Borg, & Engstrom, 2016), and to get feedback and formal acceptance from the customer. The deliverables of the sprints, such as working software modules, were used not just for acceptance, but also as the inputs for project progress monitoring and measurement (Gustavsson & Hallin, 2014; Hoda & Murgesan, 2016). Although some evidence suggested that agile approaches emphasize the use of qualitative measures in monitoring (Jahr, 2014; Misra et al., 2009), the papers also described how structured practices (Conforto & Amaral, 2010), such as explicit

Table 3.

Key Agile Goal-Setting Practices in the Papers

Agile project governance practice	Primary Studies
A. Customer-team co-operation for initial and flexible requirements and team-level goals	A. (Augustine et al., 2005; Berger & Beynon-Davies, 2009; Bjarnason et al., 2012; Conforto et al., 2014; Hoda & Murugesan, 2016; Misra et al., 2009; Ramesh et al., 2010)
B. Product backlog and vision guide prioritization and iteration process	B. (Augustine et al., 2005; Conforto & Amaral, 2010; Conforto et al., 2014; Drury-Grogan, 2014; Hoda & Murugesan, 2016; Strode et al., 2012; Torrecilla-Salinas et al., 2015)
C. User story, preferably augmented with business value perspective, should be used as the deliverable definition	C. (Bjarnason et al., 2012; Daneva et al., 2013; Strode et al., 2012; Torrecilla-Salinas et al., 2015; Trkman et al., 2016)
D. Flexible budgeting (short-term) and contracting (Time & Material) support agile projects	D. (Ben-David et al., 2012; Conboy & Morgan, 2011; Jorgensen, 2016; Misra et al., 2009)

Key Agile Incentives Practices in the Papers

Agile Project Governance Practice	Primary Studies
A. The philosophy of agile management serves as the best incentive in getting team members' commitment	A. (Chow & Cao, 2008)
B. Peer recognition, project-based organizational structures, decision-making authority, and customer contacts support the establishment of team dedication	B. (Conforto et al., 2014b; Sheffield & Lemetayer, 2013; Tessem, 2014)
C. The role of monetary incentives and risk and opportunity-sharing schemes is only limitedly addressed in motivating the individuals and organizations, and risk management is approached in a traditional manner	C. (Bjarnason et al., 2012; Conboy & Morgan, 2011; Conforto & Amaral, 2010; Shrivastava & Rathod, 2015)

Key Agile Monitoring Practices in the Papers

Agile Project Governance Practice	Primary Studies
A. Monitoring is mainly done by the team, using sprint and iteration reviews with customer feedback focus	A. (Chow & Cao, 2008; Conforto et al., 2014b; Hoda & Murugesan, 2016; Jahr, 2014; Moe et al., 2010; Ramesh et al., 2010)
B. Measures are standardized and quantitative, and agile specific and qualitative	B. (Conforto et al., 2014b; Gustavsson & Hallin, 2014; Misra et al., 2009; Persson et al., 2012; Ramesh et al., 2012; Torrecilla-Salinas et al., 2015)
C. Visual tracking project progress through deliverables	C. (Augustine et al., 2005; Miranda & Bourque, 2010; Misra et al., 2009)
D. Testing is important to validate deliverables and users' stories, and is automated if possible	D. (Bjarnason et al., 2016; Heikkila et al., 2015; Tessem, 2014; Trkman et al., 2016)

Key Agile Coordination Practices in the Papers

Agile Project Governance Practice	Primary Studies
A. Real-time, informal communication mechanisms and removing communication barriers in and between empowered teams	A. (Augustine et al., 2005; Bjarnason et al., 2016, 2012; Conforto et al., 2014b; Daneva et al., 2013; Misra et al., 2009; Persson et al., 2012; Shrivastava & Rathod, 2015)
B. Iterative project planning using product vision and backlogs	B. (Conforto & Amaral, 2010; Moe et al., 2010; Torrecilla-Salinas et al., 2015)
C. Ensuring infrastructure and practices for customer involvement and team autonomy	C. (Berger & Beynon-Davies, 2009; Chow & Cao, 2008; Conforto & Amaral, 2010; Drury-Grogan, 2014; Heikkila et al., 2015; Jahr, 2014; Moe et al., 2010; Tessem, 2014; Torrecilla-Salinas et al., 2015)
D. Managing change through continuous prioritizations	D. (Batra, 2009; Bjarnason et al., 2016; Ramesh et al., 2010, 2012)
E. Facilitating mechanisms for frequent customer deliveries	E. (Jorgensen, 2016; Stettina & Horz, 2015)

(continued)

Table 3. (continued)

Key Agile Roles and Decision-Making Power Practices in the Papers

Agile Project Governance Practice	Primary Studies
A. Agile project team with cross-functional roles and customer involvement	A. (Batra, 2009; Conboy & Morgan, 2011; Jorgensen, 2016; Shrivastava & Rathod, 2015; Tessem, 2014; Wale-Kolade, 2015)
B. Agile project team has total autonomy in the decision-making process	B. (Berger & Beynon-Davies, 2009; Chow & Cao, 2008; Conboy & Morgan, 2011; Conforto & Amaral, 2010; Conforto et al., 2014b; Drury-Grogan, 2014; Persson et al., 2012)
C. Adaptive leadership by the project manager, i.e., administrator or coordinator of the agile project team	C. (Augustine et al., 2005; McAvoy & Butler, 2007; van Waardenburg & van Vliet, 2013)
D. Agile coach oversees agile capabilities, scrum master manages sprints and project team performance	D. (Conboy & Morgan, 2011; Hoda & Murugesan, 2016)

Key Agile Capability Building Practices in the Papers

Agile Project Governance Practice	Primary Studies
A. Clients' capabilities and their establishment are the key to ensuring agile project success	A. (Bjarnason et al., 2012; Jorgensen, 2016)
B. The optimal capability composition of an agile team needs to address the tensions between high- and low-skilled workers and specialization versus cross-functionality logic	B. (Berger & Beynon-Davies, 2009; Chow & Cao, 2008; Conforto et al., 2016; Drury-Grogan, 2014; Hoda & Murugesan, 2016; Serrador & Pinto, 2015; Strode et al., 2012; Tessem, 2014; Trkman et al., 2016; Van Waardenburg & van Vliet, 2013)
C. Knowledge exchange processes within the agile development team, with the key stakeholders and permanent organization are central in capability building	C. (Bjarnason et al., 2012; Conboy & Morgan, 2011; Daneva et al., 2013; Ghobadi & Mathiassen, 2015)
D. Agile practices and tools support routinized and continuous learning	D. (Batra, 2009; Conforto & Amaral, 2010; Conforto et al., 2014b; Drury et al., 2012; Gustavsson & Hallin, 2014; McAvoy & Butler, 2007; Mistra et al., 2009; Torrecilla-Salinas et al., 2015)

definition of done (DOD) (Moe, Aurum, & Dyba, 2012), appropriate formal control (Ramesh, Mohan, & Cao, 2012), and standardized measures (Gustavsson & Hallin, 2014) can support agile development. Agile-specific metrics and key performance indicators (KPIs) that would enable monitoring in practice were seldom discussed, being addressed in only four articles. The metrics mentioned, such as the earned added value (EAV) and the agile triangle, focused mostly on customer-centric speed and value (Conforto & Amaral, 2010; Torrecilla-Salinas et al., 2015); almost equally significant were team performance-related KPIs, such as the schedule performance index (SPI) and Project Heartbeat (Augustine et al., 2005; Torrecilla-Salinas et al., 2015), that would fit the empowered team approach (Hoda & Murugesan, 2016). Visual methods, such as different regression models and burn-down charts, were described as feasible monitoring tools (Miranda & Bourque, 2010; Misra et al., 2009). Reporting, a common monitoring and controlling practice, was mentioned and described explicitly in only one paper (Torrecilla-Salinas et al., 2015).

Coordination

The literature analysis indicated several interesting patterns in and implications for agile project coordination. The main coordination practices found in the papers illustrated the methods

and tools used for iterative planning and change management activities. In agile approaches, the project plan consists of backlogs that contain the prioritized features for each sprint and user story. These backlogs and prioritizations are reviewed iteratively during each retrospective and possible changes—either from deliverable acceptance tests or updated product vision—are also revised (Bjarnason et al., 2016; Conforto & Amaral, 2010; Moe, Dingsoy, & Diba, 2010). Backlogs, and especially such sprint-level planning techniques as Kanban boards, were mostly considered as visual means of coordination rather than as formal documents, as would generally be the case in more traditional projects (Drury-Grogan, 2014; Heikkilä et al., 2015; Misra et al., 2009).

Second, the papers often addressed the role of communication—especially, but not exclusively face-to-face, visual, and informal communication—as an important coordination practice in agile projects (Augustine et al., 2005; Conforto et al., 2014b; Misra et al., 2009). In a distributed development context or a colocated geographical setting (Batra, 2009; Persson, Mathiassen, & Aen, 2012), fluent communication can be a challenge. Thus, viable solutions to these issues were also proposed in the papers: Modern IT communication and cooperation technologies, such as Slack and Skype, could support agile communication preferences by enabling

concurrent processing of deliverables and thus reducing the need for written documentation. Also, virtual presence through video streaming provides a face-to-face element that supports unspoken, informal communication even if the team is colocated (Persson et al., 2012).

In addition to the main concerns related to coordination, other topics were addressed in these papers: (1) coordination as planning changes from a periodic activity to an iterative or continuous practice as, for example, regards activities of prioritization, estimation, and release planning (Bjarnason, Wnuk, & Regnell, 2012; Bjarnason et al., 2016); (2) investigation of the likelihood of project success in relation to applied project coordination approaches (Chow & Cao, 2008); (3) open innovation in relation to agile project coordination (Conboy & Morgan, 2011); (4) construction of the agility concept, or conceptualizing agility (Conforto, Amaral, da Silva, Di Felippo, & Kamikawachi, 2016); (5) identification of challenges and efficient practices for knowledge sharing in agile teams and projects (Batra, 2009; Ghobadi & Mathiassen, 2016; Tessem, 2014); (6) empowerment and self-organizing teams in relation to agile project coordination (Tessem, 2014); (7) application of agile and traditional approaches mixed together (Jahr, 2014; van Valkenhoef, Tervonen, de Brock, & Postmus, 2011); and (8) addressing usability-related work and coordination in agile projects (Wale-Kolade, 2015). From a project governance perspective, these issues suggest that coordination practices could be grouped into three categories: (1) strategic organizational practices that take place before, after, and above the project itself, such as the facilitation of knowledge sharing that connects strongly with capability building; (2) managerial coordination practices applied by the project team within the project itself; and (3) technical coordination practices that include both agile development methods and technological tools and solutions.

Roles and Decision-Making Power

Different organizational structures are reported in the literature analyzed, suggesting that many different roles exist. One common practice is adopting an organizational structure that provides total autonomy in decision making for the agile project team (Berger & Beynon-Davies, 2009; Chow & Cao, 2008; Conboy & Morgan, 2011; Conforto & Amaral, 2010; Conforto et al., 2014b; Drury-Grogan, 2014). Often, decisions made by the team are about what to develop and when, aiming at customer satisfaction, improved project results, and team development (Bjarnason et al., 2012; Drury, Conboy, & Power, 2012; Moe et al., 2012). Such decisions are made at the group rather than the individual level and require discussion and agreement from all members (Drury et al., 2012; Maruping, Venkatesh, & Agarwal, 2009a; Moe et al., 2010). Stettina and Horz (2015) identify the autonomy of the agile team as the act of taking over many project management tasks.

Typically, the agile project team comprises different roles, including the customer, a cross-functional development team (product owner, developers, testers, architects, usability designers, etc.), and an administrative project manager

performing activities such as project planning (Conforto et al., 2014b) and deciding on requirements (Daneva et al., 2013). Increased involvement of and interactions with the customer—sometimes through a customer representative or a member from internal business line organization—with the development team are particularly distinct and highly emphasized (Batra, 2009; Conboy & Morgan, 2011; Jorgensen, 2016; Shrivastava & Rathod, 2015; Tessem, 2014). Other roles, such as those of usability designers, are also considered important but might involve the use of several tactics for full integration in the agile team (Wale-Kolade, 2015). With the expectation that the customer is competent (Jorgensen, 2016), the frequent involvement and interactions of customers with the development team facilitate intrinsic project management decisions and discussion of any changes in requirements (Conboy & Morgan, 2011; Ramesh, Cao, & Baskerville, 2010). The team relies strongly on informal roles and relationships, mutual trust, and involving the right people in the decision-making process (Persson et al., 2012). Many papers (Augustine et al., 2005; McAvoy & Butler, 2007; van Waardenburg & van Vliet, 2013) identify changes in the project manager role, adopting a new style of facilitation or coordination as they leave team members with greater autonomy to determine how tasks ought to be done while demanding demonstrable results at the end of an iteration. Outside the project team, the project management office (PMO) was also assigned an important role (Daneva et al., 2013), responsible for ensuring that successful project practices are incorporated in corporate learning practices, as ways of getting things done in future projects.

Participating in empowered decision making by the development team (Berger & Beynon-Davies, 2009; Moe et al., 2010) and adaptive leadership by the project manager (Conboy & Morgan, 2011) are reported to be challenging innovations. Members of the development team may be reluctant to make a decision for fear of taking responsibility, accountability, and blame (Berger & Beynon-Davies, 2009). An established organizational culture of passing on decision-making responsibility, often evident in hierarchical organizations, may account for such challenges (Berger & Beynon-Davies, 2009). Project-oriented matrix structures are therefore considered the best conditions for implementing the agile approach to project management (Conforto et al., 2014b; Strode, Huff, Hope, & Link, 2012). Other factors hindering group decision making include specialization and lack of training within the project team (Moe et al., 2012). Regarding adaptive leadership, according to Conboy and Morgan (2011), a project manager's workload may seem to increase in the context of autonomous teams, as a result of risks and increased requirements for coordination. This can be minimized by accepting the inability of project owners to know everything in advance while relinquishing some control to achieve greater order (Augustine et al., 2005; Maruping et al., 2009a; van Waardenburg & van Vliet, 2013).

Capability Building

The body of research analyzed in this study features multiple perspectives on capability building in agile projects, of which

the most dominant are (1) the role of client capabilities in agile projects, (2) the optimal capability composition of the agile project team and its various capability needs, (3) the processes of knowledge exchange within the agile project team and with other stakeholders and related knowledge transfer capabilities, (4) knowledge exchange and capability building across agile projects within organizations, (5) the learning and knowledge-sharing practices and tools within agile projects, and (6) the role of the agile team's experience and capabilities in explaining agile project performance. Most of the researchers in this field emphasize the important role that agile practices, such as continuous learning and end-of-iteration reflection on how to become more effective, play in capability building and the success of agile projects (Drury et al., 2012). Batra (2009), however, also highlights that although some agile teams may engage in routinized learning practices and easily adopt a continuous learning approach, for other actors this adaptive mindset might be highly challenging—for example, because of cultural reasons. The team's competence and skill structures also receive wide attention: Batra (2009) brings up tensions between high- and low-skilled team members, Tessem (2014) notes the role of experienced developers as more active decision-makers, Berger and Beynon-Davies (2009) discuss the varying degrees of competence, Drury-Grogan (2014) argues that agile members must discuss their capabilities and skills openly and honestly to divide tasks within the team wisely, and Bjarnason et al. (2012) and Jorgensen (2016) highlight the importance of stakeholders' and customers' competences and capabilities, which should include a combination of business and technical skills. Of importance is the discourse on the specialization logic of organizing the team versus the cross-functionality logic, where members aim to develop diverse skills. Conforto et al. (2016) define agility as the team's ability to change its project plans quickly in response to customer or stakeholder needs; they thus view agility as a collective capability at the agile team level. The evidence concerning the implications of the team's capabilities and experience for agile project performance has been contradictory: Although Misra et al. (2009) identified a correlation between training and learning and success, Serrador and Pinto (2015) did not find evidence for a team's experience to moderate the empirical relationship between agility and project success. Knowledge exchange and the team's capacity to codify and transfer knowledge have also received attention (Ghobadi & Mathiassen, 2016); recently, more researchers have addressed organizational-level capabilities to replicate learning from agile projects and to transfer knowledge across a project portfolio—for example, through organizational arrangements such as project management offices (PMOs) (Daneva et al., 2013).

Discussion

The aim of this study was to deepen the understanding of agile project governance by analyzing the governance-related practices used in agile projects. To achieve this goal, its results have

been elaborated by contrasting traditional and agile project governance practices in terms of which traditional project governance practices apply to agile projects as such, which need to be modified for agile contexts, and what new project governance-related practices have emerged from agile project environments. The analysis has also revealed elements and aspects of project governance about which we do not know enough, even after this study; these issues in particular would benefit from further study.

The results of this study indicate that the established governance-related practices used in agile projects, according to these papers, are connected mostly with the goal-setting, coordinating, and roles and decision-making dimensions of the governance framework proposed by Kujala et al. (2016). The study also provides insights into monitoring and capability building-related activities in agile projects, although little evidence of practices related to incentives was presented.

The study also identified many ambiguous, cross-functional, and generic governance practices in agile projects that challenged the applicability of the strict, predetermined dimensions of governance. For example, the agile practice of integrating the customer as a product owner or as a member of the project team (Berger & Beynon-Davies, 2009; Conboy & Morgan, 2011) has implications for the goal-setting dimension, through product vision and deliverable determination (Augustine et al., 2005; Strode et al., 2012); the monitoring dimension, through immediate testing and acceptance of deliverables (Jahr, 2014; Tessem, 2014); and the coordination dimension, through deliverable feedback facilitation and shared decision-making practices (Chow & Cao, 2008; Drury-Grogan, 2014). Similarly, the agile practice of building empowered, versatile project teams can be discussed under the roles and decision-making power dimension via team autonomy (Conforto & Amaral, 2010; Persson et al., 2012) or as capability building based on its emphasis on enhancing multitiered competence development (Trkman, Mendling, & Krisper, 2016; van Waardenburg & van Vliet, 2013). Although these practices reflect the flexible nature of agile development as described in the 12 principles of the *Agile Manifesto* (Beck et al., 2001), they implied connections among and therefore discussion under more than one of the six dimensions used.

Traditional Project Governance Practices That Can Be Applied to Agile as Such

Many goal-setting practices can be applied in a relatively similar manner to both traditional, plan-control-driven projects, and flexible agile projects. Formal compliance and assurance mechanisms, such as contracts and budgeting (McGrath & Whitty, 2015; Müller, 2009), are present in both contexts, even though the content, scope, and nature of the latter might focus on and enable flexibility—such as time-and-material-based contracts (Ben-David, Gelbard, & Milstein, 2012) and continuous budgeting (Conboy & Morgan, 2011). For traditional projects, goal setting is characteristically the focus of the pre-

project or front-end phase (Samset & Volden, 2016), while in agile projects this explicitly identified phase is rendered (if applied at all) as lightweight and as “visionary” as possible (Conforto et al., 2014b; Torrecilla-Salinas et al., 2015). Finally, traditional project governance also has a longer post-project perspective that is required to evaluate the long-term, strategic benefits achieved by the project (Sanderson, 2012), whereas in agile projects the perspective is distinctly short term: The focus is to complete one sprint or user story at a time and immediately to validate its benefit in terms of customer acceptance (Bjarnason et al., 2012; Jorgensen, 2016). In plan-control-driven project settings, incentives such as monetary rewards and risk and change management are elemental governance issues to be agreed upon in the pre-project phase, to ensure consensus and motivation among project parties and actors (Sanderson, 2012; Zwikael & Smyrk, 2015). In agile projects, risk management is perceived as a higher-level activity beyond the project itself. Moreover, risks are welcomed at the project level because of the acceptance of uncertainty and constant change, and are thereby not seen as negative drivers (Shrivastava & Rathod, 2015).

Traditionally, project management relies on operational, tangible measures centered on time, cost, and financial return to determine the success of a project, whereas project governance perceives success at a long-term, strategic level by validating indirect, social benefits of project outcomes (Samset & Volden, 2016; Sanderson, 2012). This study clearly indicates that even though agile projects themselves favor measuring and visualizing success and outcomes through short-term metrics and deliverables, the agile project’s long-term benefits may also be of interest for the customer, sponsor, or owner organization, and should therefore be validated with means closer to those of traditional project governance.

One central purpose of project governance is to manage and include the interests of project stakeholders, such as shareholders, financiers, suppliers, legislative bodies, nongovernmental organizations, and citizens, when constructing project objectives (Heravi, Coffey, & Trigunarsyah, 2015; Too & Weaver, 2013). The reviewed agile literature addresses mainly the internal project stakeholders that are part of a project’s daily activities (Ghobadi & Mathiassen, 2016; Heikkilä et al., 2015), while the value-adding role of external stakeholders has received limited attention. Hence, the management of external stakeholders should be considered by a governance body that oversees the project itself, similarly in both agile and traditional projects. Other external actors crucial to project governance include auditors. Traditionally, project organizations engage the assistance of external experts, such as audit bodies or committees, to ensure project management compliance and to assess and monitor project performance (Müller, 2009), whereas in agile projects the concerns about compliance are mostly related to technical aspects of software development and performance is monitored internally and in real time in sprint reviews (Miranda & Bourque, 2010; Ramesh et al., 2010).

Granting appropriate decision-making power to—or empowering—the project team is managed in traditional

project governance through ownership and resource allocation (Andersen, 2015; Winch, 2014). This generally applies in agile project teams as well (Sheffield & Lemetayer, 2013; Tessem & Maurer, 2007). Traditional project governance suggests formalized organizational structures and mechanisms for project empowerment—namely, steering groups, project portfolios, and programs (Guo, Chang-Richards, Wilkinson, & Li, 2014; McAvoy & Butler, 2009). Governance bodies external to the project, such as the PMO (Unger, Gemünden, & Aubry, 2012), must ensure that all necessary project management tasks are assigned and conducted properly. Capability and competence building are generally managed by the human resources functions of an organization, although traditional project governance suggests that a separate PMO can be established for project-related competence development and ensuring compliance (Müller, 2009; Too & Weaver, 2013). Agile software development projects require competences and know-how that are not featured among traditional project management skills, such as those related to agile practices and related technical skills (Gustavsson & Hallin, 2014; Moe et al., 2010). Consequently, a governance body external to the project should also pay close attention to extracting and consolidating the technical perspectives of customers and insights of the developers—and also pay attention to emerging trends in the community and in academia—to ensure the highest-level technical capability. Fluent sharing of knowledge (not limited to this technical capability) is also fundamental to enabling an organization’s transformation toward agility (Dove, 1999).

Traditional Project Governance Practices That Need to Be Modified for Agile

As mentioned earlier, although agile projects might have a separate front-end phase, setting the scope and goals of projects is not performed exclusively at this time, but rather is a continuous and iterative process. This is because of the accepted fact that project requirements tend to be ambiguous and change due to this uncertainty, which is not considered a limitation challenge but rather a characteristic of the environment (Drury-Grogan, 2014; Heikkilä et al., 2015).

A project plan (as a concrete document) is traditionally the main coordination and management tool that the appointed project manager completes periodically to communicate the project’s progress with stakeholders and the owning organization (Project Management Institute, 2013; Too & Weaver, 2013). However, in agile projects, the focus of the coordination and communication is to provide sufficient real-time information about the project’s status to the empowered project team and involved customer representatives visually and in real time. For this purpose, agile projects prefer product vision and backlogs instead of a formal, rigid project plan (Batra, 2009; Misra et al., 2009; Moe et al., 2012; Sheffield & Lemetayer, 2013).

Traditionally, incentives related to project performance are determined by connecting the predetermined objectives—

usually centered on time, cost, and scope—with the actual outcome of the project once the project has ended. However, in agile projects, the project objectives, as discussed in the previous section, are visionary at the best (Augustine et al., 2005) and are therefore not suitable for the measurement of success or the determination of outcome-based incentives. In agile projects, the content and targets of sprint deliverables are explicit, described in user stories that are taken from the prioritized backlog (Daneva et al., 2013). Therefore, such deliverables are used to measure and monitor a project as it progresses. The applicable indicators in agile projects thus reflect project performance (Drury-Grogan, 2014), but differ from traditional approaches by not focusing on individual tasks, activities, and people (Conforto et al., 2016; Hoda & Murugesan, 2016). Motivation in agile projects is strongly connected with the shared commitment and performance capability of the project team itself, which are characteristically inherent in agile approaches and principles per se. Externally provided incentives are not considered critical to project or individual performance. Agile projects rely on peer recognition and on rewards related to team performance as the most relevant incentives to motivation (Sheffield & Lemetayer, 2013; Tessem, 2014). Individual developers are also motivated strongly by the empowering, flexible, and trust-based approach offered by agile approaches (Chow & Cao, 2008), and do not necessarily perceive individual recognition or rewards, such as a personal performance bonus, as critical elements (Bjarnason et al., 2012; Conboy & Morgan, 2011).

Capability building in an agile project requires competence validation and development before and outside the project (Serrador & Pinto, 2015; Strode et al., 2012), which is common practice in any organization, from a project governance perspective (Biesenthal & Wilden, 2014; Janssen & van der Voort, 2016; Müller & Lecoivre, 2014). Where agile projects are distinguished is the in-project, iterative competence development, and knowledge sharing that takes place during retrospective sessions after each sprint or iteration (Batra, 2009; McAvoy & Butler, 2007).

Emerging Agile-Specific Practices With No Relationship to Traditional Project Governance

Customer involvement is not a feature exclusive to agile projects, but in agile projects it is emphasized by both the inclusion of the product owner as directly representing the customer and by ensuring that the development team members have sufficient competence in the customer business, or domain. These elements enable the formulation of iterative requirements and product vision/backlog definitions, as well as user story creation, acceptance, and validation of results (Augustine et al., 2005; Berger & Beynon-Davies, 2009; Bjarnason et al., 2012; Conboy & Morgan, 2011). The exclusive focus on the customer as the most critical goal-setting actor in agile projects can be perceived narrowly, as the product owner can subjectively determine what is needed or not. This can be challenging, especially in the context of public sector agile projects (Nuottila, Aaltonen, & Kujala, 2016).

Changes are traditionally seen as a significant challenge to project performance and success, especially in complex projects (Guo et al., 2014), whereas in characteristically uncertain agile projects, changes arise constantly in real time and are therefore not considered actual risks (Strode et al., 2012). However, if the agile team and project are distributed geographically, there are elevated risks that the project should manage more carefully, but these are not related to change as much as to communication and cultural issues (Shrivastava & Rathod, 2015). As described earlier, in agile projects, change is addressed with iterative planning and prioritization, using real-time communication methods (Augustine et al., 2005; Conforto et al., 2016; Stettina & Horz, 2015).

In their emerging project governance practice, agile projects focus on sprint deliverables as a basis for project monitoring, controlling, and incentivization. The role and criticality of these deliverables are evident in two ways. First and foremost, the predefined functional requirements and objectives of each user story and sprint are validated by deliverables through testing and customer acceptance: The product has no flaws and does what it was supposed to do (Chow & Cao, 2008; Maruping et al., 2009a). Testing—using as automated a procedure as possible—and obtaining customer acceptance should be performed immediately after the sprint is finished to avoid both work-in-progress and monitoring distortion (Bjarnason et al., 2016; Tessem, 2014; Trkman et al., 2016). Second, as mentioned before, the deliverables are the preferred method to measure the team's performance and project progress, and also form the basis for the team-level incentive system (Miranda & Bourque, 2010; Torrecilla-Salinas et al., 2015).

Agile philosophy and principles themselves can provide a solid foundation for project culture and values (Chow & Cao, 2008), because individual practitioners embrace and commit to these values consciously as they adopt agile approaches. This can be considered an emerging method for achieving a shared project culture, which has been considered a top-down organizational activity in more traditional discussions of project governance. Besides traditional project skills, a competent agile team member also requires the capability to absorb the agile philosophy and convert this into an agile way of doing (Hoda & Murugesan, 2016; Moe et al., 2010). Another emerging dimension of competence consists in the technical skills required to develop the customer's desired solution, which are generally strongly associated with the technological preferences of the customer. These preferences are not as much available to individual team members—or organizational competence-development functions such as HR or PMO—as they are to customer-facing agents, such as customer account or business units (Maruping, Zhang, & Venkatesh, 2009b). The study showed that, when the customer product owner was closely integrated, teams and individual members were able to harness technical requirements and share knowledge across teams (Bjarnason et al., 2012; Ghobadi & Mathiassen, 2016).

Roles and decision-making practices in agile projects focus almost exclusively on the construction of empowered, self-

organizing, multitasking, and committed teams. This practice is connected directly with people-related agile principles (Beck et al., 2001) and determines the team's ability to define with the customer what needs to be done before and during the project, how to validate and measure its outcomes and processes, and the capability to react and adjust to inevitable changes with fast decision-making processes (Drury-Grogan, 2014; Hoda & Murugesan, 2016; Strode et al., 2012; Tessem, 2014). However, individuals in the agile project team need appropriate authority and power to enable fast decision making in a flexible project environment (Berger & Beynon-Davies, 2009; Conforto & Amaral, 2010; Misra et al., 2009). The formal definition of individual roles and responsibilities is not deemed to be critical in agile projects, as the preference is to focus on the team with cross-functional, ambidextrous, and experienced members (Drury-Grogan, 2014; Hoda & Murugesan, 2016), which is more a capability-building topic. The only explicitly required project role is that of the customer (Conboy & Morgan, 2010; Hoda, Noble, & Marshall, 2011). The role of the traditional project manager was perceived both as a mere administrator (Augustine et al., 2005) and as an adaptive leader in relation to the team and project (McAvoy & Butler, 2007; Tessem, 2014). In the articles analyzed, the scrum master and agile coach managed the daily and sprint-level project team performance and delivery capability, respectively (Ghobadi & Mathiassen, 2016; Misra et al., 2009). Other, traditional project management-related tasks in an agile project are considered a shared responsibility of the team itself (Berger & Beynon-Davies, 2009; Conboy & Morgan, 2011). These emerging new roles and shared team responsibilities imply that the role of the traditional project manager is less important for and within the project. However, the project manager may still serve as a crucial mediator between the project and the parent organization. This is especially important in more hierarchical and bureaucratic project settings, such as in the public sector (Berger, 2007). Although the papers examined did not provide evidence of integrating the skill sets required in these roles—agile coach, scrum master, or “agile project manager”—into the PMO, research introducing the concept of an “agile PMO” does illustrate the emergence of this issue (Tengshe & Noble, 2007).

Remaining Gaps Between Agile and Traditional Project Governance

The short-term perspective adopted by agile projects toward measurement of project output, success, and benefits implies a weaker link to parent organization strategies than seen in more traditional projects. For example, connecting the initial product and project vision (Conforto & Amaral, 2010; Hoda & Murugesan, 2016) with a longer-term project outcome, and benefit validation for post-project incentives, is an interesting option that the papers in our sample do not address. This weak strategic link is also apparent in project product types: Agile approaches and projects are applied in circumstances where the product itself or the market potential is unclear and smaller in scale, whereas a traditional project approach is more common in situations where

the product and outcome are known and its business impact is significant, such as implementing a nationwide healthcare system (Conforto et al., 2014b). What remains as a gap in knowledge, after examining this body of research, is how strategic connections can be established, managed, and understood at different levels in an organization that executes agile development projects, and how the strategic alignment of objectives can be achieved; for example, arching from the project level down to individual sprints and user stories.

However, one paper in the sample did discuss practices related to strategic alignment in the agile context, but only at a portfolio level (Stettina & Horz, 2015). A project portfolio is essentially a permanent governance structure that, besides aligning and prioritizing projects within a company, is used to identify projects and allocate resources across them (Kaiser, El Arbi, & Ahlemann, 2015; Kock, Heising, & Gemünden, 2016). Another organizational-level practice that impacts goal setting of a single project is the use of programs, which consolidate projects according to their long-term or strategic business change objectives within a temporary structure, justifying the need for a more structured approach to governance and organization (Unger et al., 2012; Winch, 2014). How portfolios or programs influenced governance in agile projects remained unclear in this study, an issue that has also arisen in recent research on large agile projects in large organizations (Hobbs & Petit, 2017).

Incentives and risks are traditionally agreed upon explicitly in the project and allocated between internal or external parties. Increasingly popular project contract and delivery models, such as alliances, are elaborating the concept of joint or integrated project deliveries, where external and internal parties form a joint organization and share benefits and risks equally, for the good of the project itself (Guo et al., 2014; Lahdenperä, 2012). How agile approaches support or detract these agreements at a project level—and vice versa—from a project governance perspective remains to be researched, as such examples were not discussed in the studies reviewed. Alliances are most common in the construction industry, for large and complex projects (Lahdenperä, 2012), whereas agile approaches are most popular in small-scale projects in the software industry (Conforto et al., 2014b).

Possible conflicts are resolved in agile projects by initial agreement on the product vision (Drury-Grogan, 2014) and during the project via sprint retrospects (Daneva et al., 2013). This requires pre-established means to facilitate not only communication but also decision-making authority (Hoda & Murugesan, 2016; Sheffield & Lemetayer, 2013). In more traditional projects, conflicts such as legal disputes are generally caused by uncertainty and change, but are also anticipated during contract negotiations by both parties. Such conflicts are also usually managed outside the project (Morris & Hough, 1987; Pinto, 2014). Agile approaches challenge this arrangement by involving both project contract parties within the project team and by accepting the fact that change is inevitable and part of the working environment. The papers analyzed in this study did not provide evidence about what issues are even considered conflicts in an agile context, if and how they are

escalated, how they are managed, and how they might impact project performance. This is clearly an avenue that warrants further research.

The papers reviewed emphasized the importance of empowerment and team autonomy. However, even after this research, key questions remain open, including what degree of empowerment is appropriate and how it can be ensured in practice. Allocating power through money—by budgeting, for example—is traditionally one of the most obvious and effective methods of empowerment. However, the results of our analysis, and other research, do not suggest clear best practices for continuous budgeting, even though some commercial frameworks such as Scaled Agile (Scaled Agile Inc., 2016) are beginning to take this issue into consideration.

Our study also did not provide evidence regarding how agile projects ensure the availability of competent resources. It is not enough to have the appropriate agile and technical skills within an organization, if the resources possessing them are not available for projects. Delivery capability consists of both competence (skills and know-how) and capacity (availability), and if either aspect fails, project performance is in danger. Conflict for resources among projects is not exclusive to agile contexts, as the challenge of acquiring resources from line or business functions for project work is a very mature topic in traditional project governance literature (McGrath & Whitty, 2015; Winch, 2014). However, the uncertainty and flexibility associated with agile approaches increases the difficulty of this problem (Drury et al., 2012). Therefore, it is apparent that enterprises with a traditional, hierarchal approach to project decision making and organizational structures (McGrath & Whitty, 2015; Müller, Pemsel, & Shao, 2014) will face challenges in adapting to agile projects (Moe et al., 2012; van Waardenburg & van Vliet, 2013).

Conclusions

In this study, agile project governance was conceptualized through a systematic review of selected, high-quality research articles on agile projects, using a predefined project governance framework by Kujala et al. (2016) to identify and describe the governance practices found in the papers. Agile project governance practices take place within the project itself and focus on flexible requirements, tangible definition and validation of deliverables, iterative processes, high levels of communication, and team empowerment and capability. The purpose of agile project governance practices is to ensure the team's ability to deliver working and valid software on a continuous basis, whereas the strategic and more structured governance practices that connect with the organization are often ignored on a project level. To conclude, these results imply that the performance and the success of agile projects are best supported by practices that give project teams the freedom, authority, and capability to produce tangible value to and with the customer.

Limitations of the Study

Our study has several limitations, originating in the fact that this research process was based on the literature, not on

empirical evidence. In other words, we have analyzed what other researchers have concluded from their research on agile projects instead of looking directly at the agile projects ourselves. This approach is, however, also of value because it presents an opportunity to synthesize the existing understanding on the topic and apply extant traditional project governance concepts to it, to see where the correspondences, the tensions, and the research gaps lie. The limited number of sources and the possible biases of the researchers doing the analysis is another clear limitation. Researcher bias in the early phases of such research presents the risk that relevant papers might have been excluded, a risk evident in the number of sources whose inclusion was disputed within the research time. However, by carefully reviewing our selection criteria before and during the selection process, we were able to reach transparent and consistent decision-making processes, and were able to acquire a feasible number of journals for this study.

The scope of our study is also limited to studies done in the information systems and software development context. These studies are mostly published in outlets representing software engineering and computer science disciplines. This contextual focus is clearly another limitation to this study, as studies on project governance practices—and agile and agility, too—can be found also in other disciplines, such as in organizational and psychological discourse. Also, although sources from highly ranked journals were included, other relevant conferences papers were not. Agile practitioners prefer to publish the most recent results in conferences; however, achieving a systematic collection of conference papers was challenging because of the diversity of coverage and unclear ranking of conferences, particularly in software engineering. Thus, the sources included have been deemed to be representative, under the circumstances.

Suggestions for Future Research

Our study uncovered several interesting findings that offer the basis for a future research agenda. First, and most important, we feel that further theorization or conceptualization based on existing studies would be of strictly limited usefulness. Therefore, we encourage fellow researchers to focus on new empirical studies in this field, albeit not necessarily to the exclusion of theory. Such studies should focus on the proposed concept of agile project governance to do the following:

Assess how agile approaches, techniques, and project governance practices are applied in different levels of organizations and to understand the pervasive impact of such measures from top management via projects to individuals. This topic could increase our understanding of the external-internal aspects of governance, and also shed light on the weak strategic link between agile projects and organizations: How can organization strategies be connected to agile projects through project portfolios, programs, PMOs, and other more structured, traditional, and “external” project governance practices?

Evaluate how project contract, model, and delivery types, such as integrated project deliveries (IPD) or alliance models or

Table 4. Comparison of Traditional and Agile Project Governance Practices

Traditional Practices That Can Be Transferred to Agile as Such	Traditional Practices That Can Be Modified to Agile	Emerging Agile-Specific Practices with No Regard to Traditional	Remaining Gaps Between Agile and Traditional Project Governance
Distinguished front-end phase for goal setting and business case	Scope setting is visionary and iterative process, starting with product vision and user stories	Inclusion of customer within the project team for decision making and validation efficiency	Linking and aligning the agile project objectives with organizations' strategy
Long-term post-project benefit realization	Short-term value realization through utilization of deliverables	Using validated—tested and accepted—deliverables as project performance and incentives basis	Connection between formal and flexible budgeting
Clear organizational and ownership structure	Flexible, value-based project budgeting and contracting	Change management is embedded into iterative planning and delivery	Impact of collaborative project models to agile project governance
External audits by third parties	Motivation through peer and team performance-related incentives	Agile philosophy and principles as project culture foundation	Conflict escalation and resolution within agile project
Project stakeholder management	Visual, real-time project monitoring through agile specific and traditional measures	Project team empowerment for autonomy, flexibility, and decision-making power	Application and impact of formalized project governance structures such as steering groups, portfolios, and programs in agile context
Project risk management	Product and sprint backlogs are the main planning and communication means	Project team composed of multitasked, self-organized, and cross-functional individuals, both experts and beginners	Project human resource and capacity management in agile
Necessary documentation for external validation and communication	Onsite, informal, and real-time communication (internal)	Agile mindset and technical capabilities as competence requirements	
Project capability management on organizational level through HR and PMOs	Project management daily activities are shared on team level	Project manager role changes to administrative and facilitative	
	Sprint retrospectives used as interproject learning method	Agile coach and scrum master manage agile project team performance	

outsourcing and offshoring, impact the application and utilization of agile approaches from a project governance perspective. What are the characteristics or prerequisites of an agile alliance? Does outsourcing support or detract the use of agile approaches—or does it impact it at all? Is anything in particular required for the governance of outsourced projects run in an agile approach?

Analyze how agile project governance scales, changes, or evolves according to project size (small or large scale), type (pure agile, hybrid, or traditional), complexity (organizational, geographical, or cultural), or product construct (“off-the-shelf” product, platform, tailored, or new product). Also, the project segment and life cycle could be considered under this area: A design phase could be completely agile, but implementation and maintenance completely traditional. How would this affect the governance of the whole project?

Understand how agile project governance is perceived and applied in different empirical (industries, private and public sectors) and academic contexts (research streams). How does (external) governance turn into (internal) governance in different contexts, and what might explain these differences and commonalities? Are there areas where agile project governance still contradicts traditional methods? Or are there areas where the new, emergent tactics have not yet been properly identified and described?

The literature on agile approaches to project management has suggested a wide array of applicable governance practices. Some of these correspond to traditional methods, while others seem better suited specifically for informal and flexible agile approaches, as presented earlier in Table 4. This study contributes to the literature by applying the underlying techniques suggested by Kujala et al. (2016) to the body of papers pertaining to agile approaches to project management and illustrating the traditional governance practices that can be applied to agile projects, the emerging practices in agile contexts that are new to traditional project governance, and the areas where gaps in knowledge remain. Second, the study systematically distills from a wide array of agile literature the portion that applies to project governance and makes this theme the subject of explicit, focused attention. The study thereby provides new knowledge for both project management and software engineering fields and encourages academics to conduct further research on the topic in other contexts as well. This study also has implications for management, by enabling agile project teams and owner organizations to identify and apply feasible governance practices within and outside the project that can support both better agile project performance and governance.

Appendix I—Selected journals

Appendix I. Source Journals.

Full Journal Title	WoS Rank	Impact Factor	SJR Rank	SJR	No. of Primary Studies	Primary Studies (Appendix 4)
<i>Academy of Management Annals</i>	1	9.74	1	15.472	—	
<i>Academy of Management Journal</i>	3	6.23	2	10.317	—	
<i>Academy of Management Perspectives</i>	16	3.94	10	4.258	—	
<i>Academy of Management Review</i>	2	7.29			—	
<i>ACM Transactions on Software Engineering and Methodology</i>	21	1.51	90	0.98	—	
<i>Administrative Science Quarterly</i>	6	5.32			—	
<i>Advances in Engineering Software</i>	14	1.67			—	
<i>Annual Reviews in Control</i>			20	2.443	—	
<i>Applied Soft Computing Journal</i>			37	1.763	—	
<i>Business Strategy and the Environment</i>	19	3.48	23	1.873	—	
<i>Communications of the ACM</i>	2	3.30			2	P1, P2
<i>IEEE Transactions on Affective Computing</i>			32	1.872	—	
<i>IEEE Transactions on Software Engineering</i>	20	1.52	52	1.543	—	
<i>Information and Software Technology</i>	16	1.57	99	0.92	15	P5, P6, P8, P14, P17, P20, P22, P25, P26, P33, P37, P38, P39, P40, P41
<i>Information Fusion</i>			31	1.941	—	
<i>Information Systems Journal</i>			51	1.575	4	P4, P15, P, 28, P29
<i>Information Systems Research</i>	24	3.05			2	P21, P30
<i>INFORMS Journal on Computing</i>			40	1.751	—	
<i>International Journal of Management Reviews</i>	7	4.85			—	
<i>International Journal of Project Management</i>	29	2.89			6	P3, P11, P16, P31, P32, P35
<i>Journal of International Business Studies</i>			12	4.208	—	
<i>Journal of International Management</i>			24	1.829	—	
<i>Journal of Management</i>	4	6.05			—	
<i>Journal of Management Studies</i>	12	4.13			—	
<i>Journal of Scientific Computing</i>			41	1.724	—	
<i>Journal of Systems and Software</i>	24	1.42	105	0.897	11	P7, P12, P14, P18, P23, P24, P27, P34, P36, P42
<i>Journal of Technology Transfer</i>			29	1.518	—	
<i>Journal of the ACM</i>	12	1.80	19	2.509	—	
<i>Knowledge-Based Systems</i>			26	2.14	—	
<i>Leadership Quarterly</i>			17	2.77	—	
<i>Management and Organization Review</i>			14	3.435	—	
<i>Omega-International Journal of Management Science</i>	15	3.96			—	
<i>Project Management Journal</i>	77	1.77			3	P9, P10, P19
<i>Research Policy</i>	20	3.47			—	
<i>Science of Computer Programming</i>	62	0.83	197	0.57	—	
<i>Strategic Management Journal</i>	21	3.38	4	6.278	—	
<i>Strategic Organization</i>			18	2.733	—	

Appendix 2. Article Search Details

1. Search string for WoS articles

SO (Publication name) = (ACADEMY OF MANAGEMENT ANNALS OR ACADEMY OF MANAGEMENT JOURNAL OR ACADEMY OF MANAGEMENT PERSPECTIVES OR ACADEMY OF MANAGEMENT REVIEW OR ACM TRANSACTIONS ON SOFTWARE ENGINEERING “AND” METHODOLOGY OR ADMINISTRATIVE SCIENCE QUARTERLY OR ADVANCES

IN ENGINEERING SOFTWARE OR ANNUAL REVIEWS IN CONTROL OR APPLIED SOFT COMPUTING OR BUSINESS STRATEGY “AND” THE ENVIRONMENT OR COMMUNICATIONS OF THE ACM OR IEEE TRANSACTIONS ON AFFECTIVE COMPUTING OR IEEE TRANSACTIONS ON SOFTWARE ENGINEERING OR INFORMATION “AND”

SOFTWARE TECHNOLOGY OR INFORMATION FUSION OR INFORMATION SYSTEMS JOURNAL OR INFORMATION SYSTEMS RESEARCH OR INFORMS JOURNAL ON COMPUTING OR INTERNATIONAL JOURNAL OF MANAGEMENT REVIEWS OR INTERNATIONAL JOURNAL OF PROJECT MANAGEMENT OR JOURNAL OF INTERNATIONAL BUSINESS STUDIES OR JOURNAL OF INTERNATIONAL MANAGEMENT OR JOURNAL OF MANAGEMENT OR JOURNAL OF MANAGEMENT STUDIES OR JOURNAL OF SCIENTIFIC COMPUTING OR JOURNAL OF SYSTEMS “AND” SOFTWARE OR JOURNAL OF TECHNOLOGY TRANSFER OR JOURNAL OF THE ACM OR KNOWLEDGE BASED SYSTEMS OR LEADERSHIP QUARTERLY OR MANAGEMENT “AND” ORGANIZATION REVIEW OR OMEGA INTERNATIONAL JOURNAL OF MANAGEMENT SCIENCE OR PROJECT MANAGEMENT JOURNAL OR RESEARCH POLICY OR SCIENCE OF COMPUTER PROGRAM-

MING OR STRATEGIC MANAGEMENT JOURNAL OR STRATEGIC ORGANIZATION)

TS (Topic) = (AGILE AND PROJECT*)”

2. document type (applied on search results) = ‘article or proceeding paper’

3. Search strings for Scopus articles

Publication title by selecting from the results the journals that were included in the list of selected journals: (LIMIT-TO (EXACTSRCTITLE, “Journal Of Systems And Software”) OR LIMIT-TO (EXACTSRCTITLE, “International Journal Of Project Management”) OR LIMIT-TO (EXACTSRCTITLE, “Project Management Journal”) OR LIMIT-TO (EXACTSRCTITLE, “Information Systems Journal”) OR LIMIT-TO (EXACTSRCTITLE, “Science Of Computer Programming”))

Document type: (LIMIT-TO (DOCTYPE, “ar”) OR LIMIT-TO (DOCTYPE, “cp”) OR LIMIT-TO (DOCTYPE, “ip”))

Appendix 3—Inclusion details

Appendix 3. Inclusion Details per Step.

Database	Inclusion/Exclusion	STEP 1			STEP 2				
		R1	R2	R3	R1	R2	R3	Decision	Discrepancies
WoS	Yes (Include)	57	32	57	56	56	55	43	
	No (Exclude)	53	78	53	54	54	55	40	
	TOTAL	110	110	110	110	110	110	83	27
SCOPUS	Yes (Include)	3	7	8	3	7	8	1	
	No (Exclude)	52	48	47	52	48	47	41	
	TOTAL	55	55	55	55	55	55	42	13
WoS + SCOPUS	Yes (Include)	60	39	65	59	63	63	43	
	No (Exclude)	105	126	100	106	102	102	82	
	TOTAL	165	165	165	165	165	165	125	40

Appendix 4—Primary studies with full citations

Appendix 4. Primary Studies with Full Citations.

Study	Citation (Full)
P1	Augustine, S., Payne, B., Sencindiver, F., & Woodcock, S. (2005). Agile project management: Steering from the edges. <i>Communications of the ACM</i> , 48(12), 85–89.
P2	Batra, D. (2009). Modified agile practices for outsourced software projects. <i>Communications of the ACM</i> , 52(9), 143–148.
P3	Ben-David, A., Gelbard, R., & Milstein, I. (2012). Supplier ranking by multi-alternative proposal analysis for agile projects. <i>International Journal of Project Management</i> , 30(6), 723–730.
P4	Berger, H., & Beynon-Davies, P. (2009). The utility of rapid application development in large-scale, complex projects. <i>Information Systems Journal</i> , 19(6), 549–570.
P5	Bjarnason, E., Unterkalmsteiner, M., Borg, M., & Engstrom, E. (2016). A multi-case study of agile requirements engineering and the use of test cases as requirements. <i>Information and Software Technology</i> , 77, 61–79.
P6	Bjarnason, E., Wnuk, K., & Regnell, B. (2012). Are you biting off more than you can chew? A case study on causes and effects of overscoping in large-scale software engineering. <i>Information and Software Technology</i> , 54(10), 1107–1124.
P7	Chow, T., & Cao, D.-B. (2008). A survey study of critical success factors in agile software projects. <i>Journal of Systems and Software</i> , 81(6), 961–971.
P8	Conboy, K., & Morgan, L. (2011). Beyond the customer: Opening the agile systems development process. <i>Information and Software Technology</i> , 53(5), 535–542.
P9	Conforto, E. C., Amaral, D. C., Carlos, E. C., & Amaral, D. C. (2010). Evaluating an agile method for planning and controlling innovative projects. <i>Project Management Journal</i> , 41(2), 73–80.
P10	Conforto, E. C., Amaral, D. C., da Silva, S. L., Di Felippo, A., & Kamikawachi, D. S. L. (2016). The agility construct on project management theory. <i>International Journal of Project Management</i> , 34(4), 660–674.
P11	Conforto, E. C., Salum, F., Amaral, D. C., da Silva, S. L., & de Almeida, L. F. (2014). Can agile project management be adopted by industries other than software development? <i>Project Management Journal</i> , 45(3), 21–34.
P12	Daneva, M., van der Veen, E., Amrit, C., Ghaisas, S., Sikkil, K., Kumar, R., ... Wieringa, R. (2013). Agile requirements prioritization in large-scale outsourced system projects: An empirical study. <i>Journal of Systems and Software</i> , 86(5), 1333–1353.
P13	Drury, M., Conboy, K., & Power, K. (2012). Obstacles to decision making in agile software development teams. <i>Journal of Systems and Software</i> , 85(6), 1239–1254.
P14	Drury-Grogan, M. L. (2014). Performance on agile teams: Relating iteration objectives and critical decisions to project management success factors. <i>Information and Software Technology</i> , 56(5), 506–515.
P15	Ghobadi, S., & Mathiassen, L. (2016). Perceived barriers to effective knowledge sharing in agile software teams. <i>Information Systems Journal</i> , 26(2), 95–125.
P16	Gustavsson, T. K., & Hallin, A. (2014). Rethinking dichotomization: A critical perspective on the use of “hard” and “soft” in project management research. <i>International Journal of Project Management</i> , 32(4), 568–577.
P17	Heikkilä, V. T., Paasivaara, M., Rautiainen, K., Lassenius, C., Toivola, T., & Jarvinen, J. (2015). Operational release planning in large-scale Scrum with multiple stakeholders—A longitudinal case study at F-secure corporation. <i>Information and Software Technology</i> , 57, 116–140.
P18	Hoda, R., & Murugesan, L. K. (2016). Multi-level agile project management challenges: A self-organizing team perspective. <i>Journal of Systems and Software</i> , 117, 245–257.
P19	Jahr, M. (2014). A hybrid approach to quantitative software project scheduling within agile frameworks. <i>Project Management Journal</i> , 45(3), 35–45.
P20	Jorgensen, M. (2016). A survey on the characteristics of projects with success in delivering client benefits. <i>Information and Software Technology</i> , 78, 83–94.
P21	Maruping, L. M., Venkatesh, V., & Agarwal, R. (2009). A control theory perspective on agile methodology use and changing user requirements. <i>Information Systems Research</i> , 20(3), 377–399.
P22	McAvoy, J., & Butler, T. (2007). The impact of the Abilene Paradox on double-loop learning in an agile team. <i>Information and Software Technology</i> , 49(6), 552–563.
P23	Miranda, E., & Bourque, P. (2010). Agile monitoring using the line of balance. <i>Journal of Systems and Software</i> , 83(7), 1205–1215.
P24	Misra, S. C., Kumar, V., & Kumar, U. (2009). Identifying some important success factors in adopting agile software development practices. <i>Journal of Systems and Software</i> , 82(11), 1869–1890.
P25	Moe, N. B., Aurum, A., & Dyba, T. (2012). Challenges of shared decision-making: A multiple case study of agile software development. <i>Information and Software Technology</i> , 54(8), 853–865.
P26	Moe, N. B., Dingsoyr, T., & Dyba, T. (2010). A teamwork model for understanding an agile team: A case study of a Scrum project. <i>Information and Software Technology</i> , 52(5), 480–491.
P27	Nurdiani, I., Borstler, J., & Fricker, S. A. (2016). The impacts of agile and lean practices on project constraints: A tertiary study. <i>Journal of Systems and Software</i> , 119, 162–183.
P28	Persson, J. S., Mathiassen, L., & Aaen, I. (2012). Agile distributed software development: enacting control through media and context. <i>Information Systems Journal</i> , 22(6), 411–433.

(continued)

Appendix 4. (continued)

Study	Citation (Full)
P29	Ramesh, B., Cao, L., & Baskerville, R. (2010). Agile requirements engineering practices and challenges: an empirical study. <i>Information Systems Journal</i> , 20(5), 449–480.
P30	Ramesh, B., Mohan, K., & Cao, L. (2012). Ambidexterity in agile distributed development: An empirical investigation. <i>Information Systems Research</i> , 23(2), 323–339.
P31	Serrador, P., & Pinto, J. K. (2015). Does agile work? A quantitative analysis of agile project success. <i>International Journal of Project Management</i> , 33(5), 1040–1051.
P32	Sheffield, J., & Lemetayer, J. (2013). Factors associated with the software development agility of successful projects. <i>International Journal of Project Management</i> , 31(3), 459–472.
P33	Shrivastava, S. V., & Rathod, U. (2015). Categorization of risk factors for distributed agile projects. <i>Information and Software Technology</i> , 58, 373–387.
P34	Stankovic, D., Nikolic, V., Djordjevic, M., & Cao, D.-B. (2013). A survey study of critical success factors in agile software projects in former Yugoslavia IT companies. <i>Journal of Systems and Software</i> , 86(6), 1663–1678.
P35	Stettina, C. J., & Horz, J. (2015). Agile portfolio management: An empirical perspective on the practice in use. <i>International Journal of Project Management</i> , 33(1), 140–152.
P36	Strode, D. E., Huff, S. L., Hope, B., & Link, S. (2012). Coordination in co-located agile software development projects. <i>Journal of Systems and Software</i> , 85(6), 1222–1238.
P37	Tessem, B. (2014). Individual empowerment of agile and non-agile software developers in small teams. <i>Information and Software Technology</i> , 56(8), 873–889.
P38	Torrecilla-Salinas, C. J., Sedenio, J., Escalona, M. J., & Mejias, M. (2015). Estimating, planning and managing agile web development projects under a value-based perspective. <i>Information and Software Technology</i> , 61, 124–144.
P39	Trkman, M., Mendling, J., & Krisper, M. (2016). Using business process models to better understand the dependencies among user stories. <i>Information and Software Technology</i> , 71, 58–76.
P40	Wale-Kolade, A. Y. (2015). Integrating usability work into a large inter-organisational agile development project: Tactics developed by usability designers. <i>Journal of Systems and Software</i> , 100, 54–66.
P41	van Waardenburg, G., & van Vliet, H. (2013). When agile meets the enterprise. <i>Information and Software Technology</i> , 55(12), 2154–2171.
P42	van Valkenhoef, G., Tervonen, T., de Brock, B., & Postmus, D. (2011). Quantitative release planning in extreme programming. <i>Information and Software Technology</i> , 53(11), 1227–1235.

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